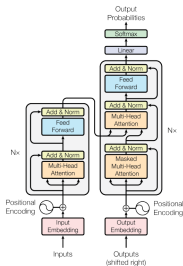


Deep Learning for NLP

Transformer

A Universal Deep Learning Architecture



Learning goals

- Understand the initial use of the Transformer
- Grasp the other application fields since then

MACHINE TRANSLATION I

- Sequence-to-sequence task
- Already served as a motivation for introducing the "ordinary" Attention-mechanism ▶ Bahdanau et al., 2015
- Crucial, that the decoder has access to the whole input sequence
→ This is very well solved by cross-attention
- Good contextualization in the encoder improves translation quality
 - (Bidirectional) RNNs/LSTMs are only (concatenated) unidirectional architectures
 - Transformer-Encoder layers are bidirectional by construction
 - Stacking them on top of each other makes this bidirectional contextualization even "deeper"

WMT 2014 EN-TO-DE AND EN-TO-FR I

Parallel training data:

Parallel data:

File	Size	CS-EN	DE-EN	HI-EN	FR-EN	RU-EN	Notes
Europarl v7	628MB	✓	✓		✓		same as previous year, corpus home page
Common Crawl corpus	876MB	✓	✓		✓	✓	same as previous year
UN corpus	2.3GB				✓		same as previous year, corpus home page
News Commentary	77MB	✓	✓		✓	✓	updated, data with document boundaries
10²French-English corpus	2.3 GB				✓		same as previous year [md5] sha1
CzEng 1.0	115MB	✓					same as previous year, corpus home page (avoid sections 98 and 99)
Yandex 1M corpus	121MB					✓	corpus home page ; v1.3 now in original case
Wiki Headlines	7.8MB			✓		✓	Provided by CMU. The ru-en is unchanged from last year.
HindEnCorp	25MB			✓			Collected by Charles University
The JHU Corpus				✗			This is fully contained in HindEnCorp, so not made available here.

THE BLEU SCORE I

- Based on n-gram overlap from candidate and reference sentence
- Precision (for each n-gram) calculated as follows:

$$P_n = \frac{\sum_{C \in \{ \text{Candidates} \}} \sum_{n\text{-gram} \in C} \text{Count}_{\text{clip}}(n\text{-gram})}{\sum_{C' \in \{ \text{Candidates} \}} \sum_{n\text{-gram}' \in C'} \text{Count}(n\text{-gram}')}$$

- Finally, the BLEU score can be computed as

$$\text{BLEU} = BP \cdot \exp \left(\sum_{n=1}^N w_n \cdot \log(P_n) \right),$$

- where BP is a "brevity penalty" to penalize short generations, N is the number of n-grams & w_n the weight for each P_n (usually $\frac{1}{N}$)

TRANSFORMER FOR MT I

The Transformer ..

- .. outperforms the previous SOTA models
- .. at a lower number of required training FLOPs

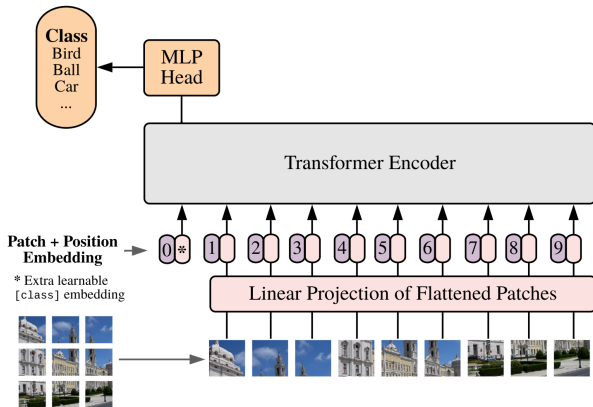
Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

► Source: Vaswani et al., 2017

TRANSFORMER FOR COMPUTER VISION I

Vision Transformer

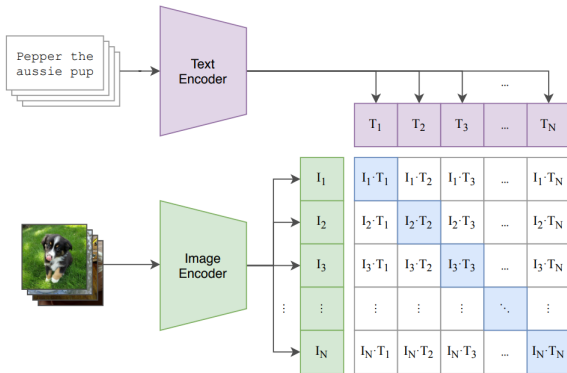


► Source: Dosovitskiy et al., 2020

TRANSFORMER FOR MULTIMODAL LEARNING I

CLIP

(1) Contrastive pre-training

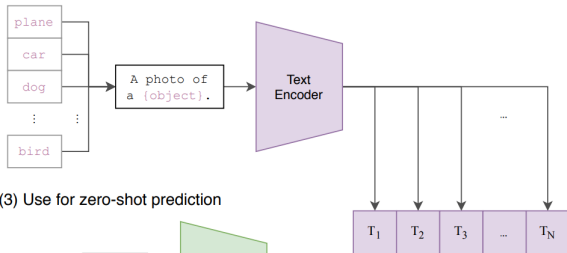


► Source: Radford et al., 2021

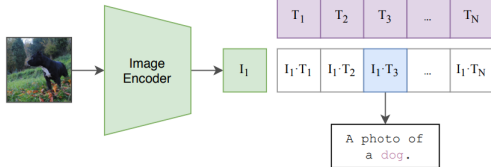
TRANSFORMER FOR MULTIMODAL LEARNING I

CLIP

(2) Create dataset classifier from label text



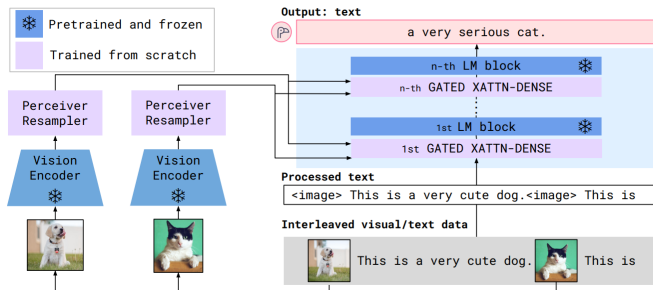
(3) Use for zero-shot prediction



► Source: Radford et al., 2021

TRANSFORMER FOR MULTIMODAL LEARNING I

FLAMINGO



► Source: Alayrac et al., 2022

TRANSFORMER FOR MULTIMODAL LEARNING I

GPT-4

Example of GPT-4 visual input:

User What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/t/1amm/comments/sab5v6/1amm/>

GPT-4 The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

Table 3. Example prompt demonstrating GPT-4's visual input capability. The prompt consists of a question about an image with multiple panels which GPT-4 is able to answer.

► Source: OpenAI, 2023